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St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001
B.Sc Degree Examination – APRIL 2021

CH616S – THERMODYNAMICS OF IDEAL AND NON IDEAL SOLUTIONS

Time : 3 hrs **Max Marks :75**

SECTION – A (20X1=20)

Answer **ALL** Questions

I. Choose the correct answer

1. Change in enthalpy during mixing of ideal solution is
   a) Negative b) Zero c) Positive d) All the above

2. Depression in freezing point depends on
   a) Molality of solvent b) Molality of solute.
   c) Normality of solute d) Normality of solvent

3. Among the following which will not affect an equilibrium constant?
   a) Temperature b) Catalyst c) Enzyme d) Both b) and c)

4. Equilibrium constant and Gibbs free energy are related as
   a) $\Delta G = RT \ln K$ b) $-\Delta G = RT \ln K$
   c) $-\Delta G = R \ln K$ d) $\Delta G = R \ln K$

5. Predict the base in the following reaction $HS^- + H_2O \leftrightarrow H_3O^+ + S^{2-}$
   a) H_2O b) S^{2-} c) Both a) and b) d) HS^-

6. What will be the solubility product of AgCl in water if its solubility is 0.00179 g per litre.
   a) 1.56×10^{-10} b) 1.56×10^{10} c) 1.56 d) 10.56×10^{-10}

7. Unit for equivalent conductance is
   a) $ohm^{-1}\ cm^2\ equi$ b) $ohm^{-1}\ cm^2\ equi^{-1}$
   c) $ohm\ cm^2\ equi^{-1}$ d) $ohm^{-1}\ cm\ equi^{-1}$

8. Calomel is
   a) Mercurous chloride b) KCl c) Hg d) All the above

9. EMF of an electrochemical cell with transference can be calculated from
   a) $t_-RT \ln a_2/a_1$ b) $t_-RT/F \ln a_2$
   c) $RT/F \ln a_2/a_1$ d) $t_-RT/F \ln a_2/a_1$

10. Liquid junction potential will be negative if
    a) $t_+ < t_-$ b) $t_+ = t_-$ c) $t_+ > t_-$ d) cannot be predicted

II. Fill up the blanks

11. At _____ composition the liquid phase solidifies without change in composition.

12. Chemical equilibrium will be _____ in nature.

13. _____ can be used as an indicator for the titration between weak acid and strong base.

14. Ratio of specific conductance to measured conductance is called _____.

15. Metal present in glass electrode is _____.

III. Match the following

16. Elevation of boiling point	a) salt bridge
17. Spontaneous process	b) H_3O^+
18. An acid	c) K_bx
19. Resistance	d) $\Delta G < 0$
20. LJP	e) l/a

SECTION – B (10X2=20)

Answer any **TEN** Questions

21. State Raoult's law.

22. Define upper critical solution temperature.

23. If the latent heat of fusion of ice is 334.7 J per gram at 0^oC, calculate the molal depression constant of water.

24. How does equilibrium respond to changes of pressure?

25. Write two limitations of chemical equilibrium.

26. What is a conjugate pair?

27. Define a buffer solution.

28. Define specific conductance.

29. How is spontaneity of a cell reaction predicted?

30. Write the electrode reaction for the cell Cd,Cd^{2+} // Cu^{2+},Cu.

31. Write the formula to calculate the emf of a concentration cell with transference.

32. Write the electrode reaction of quinhydrone electrode.

SECTION- C (5X7=35)

Answer any **FIVE** Questions

33. a) Explain the phase diagram of phenol- water system. **(4)**
    b) What are ideal and non ideal solutions? **(3)**

34. a) Derive van't Hoff equation. **(4)**
    b) Equilibrium constant (Kp) for the formation equilibrium of ammonia from nitrogen and hydrogen at $400^{o}C$ is 1.64×10^{-4} atm. Calculate the equilibrium constant at $500^{o}C$ if enthalpy of reaction is -105185.8 J. **(3)**

35. a) Explain the titration between a strong acid and strong base. **(4)**
    b) Explain Bronsted – Lowry theory. **(3)**

36. a) Write the half cell and cell reaction of Daniel cell. **(3)**
    b) A potential of 12V was applied to two electrodes placed 20cm apart. A dilute solution of ammonium chloride was placed between the electrodes when ammonium ion was found to cover a distance of 1.6 cm in one hour. Calculate the mobility of ammonium ion. **(4)**

37. a) How is ionic mobility determined? **(4)**
    b) Explain the working of chlorine electrode. **(3)**

38. a) Calculate the standard EMF and equilibrium constant for the cell $Cu,Cu^{2+}//Cl^{-},Cl_2(g),Pt$. The reduction potential of copper electrode and chlorine electrodes are 0.34V and 1.36V respectively. **(4)**
    b) Calculate the EMF of the concentration cell Zn, $Zn^{2+}(0.01m)//$ $Zn^{2+}(0.1m),Zn$. Assume thatmean activity coefficient is unity for the electrolyte. **(3)**

39. a) Derive the emf relation for an electrochemical cell without transference. **(4)**
    b) What is electrochemical series? Explain. **(3)**

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